ENN583 Foundations of Robotic Vision · exam cheatsheet

W1 Images

- img[v,u] = [row, col]; origin top-left; uint8 0–255.
- Chromaticity r = R/(R+G+B) (white ≈ 0.33).
- Central diff: $f' \approx (f(x+\varepsilon) - f(x-\varepsilon))/2\varepsilon$; I_x kernel [−½ 0 ½].
- $\nabla I = (I_x, I_y)$; $|\nabla I| = \sqrt{(I_x^2 + I_y^2)}$; $\theta = \text{atan2}(I_y, I_x) \perp$ edge. Smooth first: $\partial(I * H) = I * \partial H$.

W2 Features

- Good feature: distinctive, repeatable, well localised.
- Harris $A = w * [I_u^2 \; I_u I_v; \; I_u I_v \; I_v^2]$: λ_1, λ_2 large = corner; one large = edge; both small = flat.
- Moravec: min SSD over shifts (fails on diagonal edges).
- SIFT: DoG $D = L(k\sigma) - L(\sigma)$; extrema vs 26 neighbours; orientation histogram; 4×4×8 = 128-D.
- BRIEF: random pair tests → 256 bits; Hamming; not rot/scale invariant. ORB = oFAST (pyramid, Harris score, intensity-centroid angle) + rBRIEF.
- FAST: N ≥ 12 contiguous of 16 brighter/darker than I(p) ± t; quick test 1,5,9,13 (≥ 3 of 4).
- Ratio test $d_1/d_2 < \theta$ (strict). Cross-check = mutual NN. Euclid for float, Hamming for binary.
- Baseline: ORB + Hamming + ratio/cross-check → RANSAC. SIFT for big scale/viewpoint change.
- SuperGlue: self + cross attention GNN (9 layers) → score matrix + dustbin → Sinkhorn (OT). AUC@20 36.4 → 51.8, precision 50.4 → 84.4%. LightGlue: adaptive depth (early exit) + width (prune points); 44 ms vs 70 ms (adaptive 31 ms).

W3 Camera & stereo

- $x = fX/Z$; $u = f \cdot X/Z + u_0$ (f in px = f_mm/pixel size). $K = [f \; 0 \; u_0; \; 0 \; f \; v_0; \; 0 \; 0 \; 1]$.
- $\tilde{p} = K[R|t]P$; C 3×4, scale-free → 11 DOF. Pose (R,t) of camera → extrinsic is its inverse.
- Lost: depth, lengths, angles, parallelism. Kept: straight lines. Vanishing points/lines.
- Thin lens $1/f = 1/Z + 1/e$; blur $R = L\delta/2e$; $Z \gg f$ → pinhole.
- Bayer RGGB (2× green). Global vs rolling shutter (skew).
- **$Z = fb/d$**, $X = Z(u-u_0)/f$, $Y = Z(v-v_0)/f$; $\delta Z \approx Z^2 \delta d/(fb)$; Zmin = fb/d_max.
- Big baseline: ↓δZ, ↑Zmin, harder close search. Rectify → search same row.
- Plane → homography H (8 DOF, 4 pts, DLT + SVD).

W4 Epipolar & RANSAC

- $x'^T F x = 0$, $l' = Fx$; F rank 2, 7 DOF; 8-pt: $Af = 0$, SVD, force rank 2, normalise.
- Epipole = other camera centre's image; all epipolar lines pass through it.
- Degenerate: pure rotation, planar scene, collinear points; near: small baseline, far points, forward motion.
- $\hat{x} = K^{-1}x$; $\hat{x}'^T E \hat{x} = 0$; **$E = K'^T F K = [t]_\times R$**; 5 DOF, ≥ 5 pts; 4 (R,t) → cheirality picks 1; t only up to scale.
- **$k = \log(1-p)/\log(1-w^n)$**. w=.5,p=.99: n=8 → 1177; 5 → 146; 3 → 35. w=.9,n=8,p=.999 → 13.
- Raising threshold: old inliers stay, outliers may join.
- Variants: PROSAC, LO-RANSAC, MAGSAC++, GC-, USAC.

W5 Visual odometry

- Scale ambiguity: $K(R \cdot sX + s \cdot t) = s \cdot K(RX + t)$. Fix with known 3D: $s^* = \text{argmin} \; \Sigma \| x - \pi(K(Rp + st)) \|^2$.
- PnP: 1 pt ∞, 2 pts ∞ (bounded), **P3P ≤ 4 solutions** (+1 pt), ≥ 4 EPnP. Pipeline: P3P + RANSAC → LM refine on inliers.
- Residual = obs − pred; inlier if $\|r\| < \tau$ (norm, not squared).
- 3D–3D (Arun): centroids; $H = \Sigma \, \bar{x}_{t-1} \bar{x}_t^T$; $USV^T = \text{svd}(H)$; $R = VU^T$; $t = \bar{x}_t - R \, \bar{x}_{t-1}$. Less accurate (stereo depth error).
- **$^A T_C = {}^A T_B \; {}^B T_C$**; C_k = C_{k−1}T_k; $^B T_A = ({}^A T_B)^{-1}$.
- Keyframes: motion threshold, lost features, time; mono: baseline/avg depth > 10–20%.
- Optimisation template: parameters / prediction / residual / objective. Exact → solve; linear → QR/SVD; nonlinear → GN/LM; outliers → robust/RANSAC.

W6 Graphs & BA

- Pose graph: $X^* = \text{argmin} \; \Sigma \| x_i - {}^i T_j \, x_j \|^2_\Sigma$ (poses only, relative-pose factors). BA: $\text{argmin} \; \Sigma \| y_j^i - \pi(p_j, x_i, K) \|^2$ (poses + landmarks, reprojection).
- Weight $1/\sigma^2$: σ halves → 4× influence.
- GN: $J^T J \Delta x = -J^T r$; LM: $(J^T J + \lambda I)\Delta x = -J^T r$. Sparse normal equations. Needs a good initial guess (local minima).
- Robust: quadratic (unbounded); L1, Huber k = 1.345 (bounded); Tukey 4.685, Cauchy (redescending); Tukey/Hampel zero influence beyond cut-off.
- Factor graph: bipartite, undirected; variables + factors. Joint = Π factors.
- Gaussian fusion: $\mu = (\mu_1/\sigma_1^2 + \mu_2/\sigma_2^2)/(1/\sigma_1^2 + 1/\sigma_2^2)$, $\sigma^2 = 1/(\Sigma 1/\sigma_i^2)$.
- **MAP + Gaussian = weighted nonlinear LS** (log, drop const, Mahalanobis $e^T \Sigma^{-1} e$). MAP loses spread.

W7 SLAM / VIO / eval

- Direct: photometric error, dense/semi-dense, sub-pixel, needs good init, illumination-sensitive (DSO, DTAM, LSD). Indirect: reprojection, sparse, wide basin (ORB-SLAM, PTAM).
- IMU: gyro ω + accel a; $\tilde{a} = R(a - g) + b + n$; bias → error $\frac{1}{2}bt^2$ (0.05 m/s², 10 s → 2.5 m). Camera bounds drift; IMU gives scale, robustness, 1–10 kHz.
- ATE: align (R,T,s), RMSE of positions (global). RPE: relative-motion error (local drift). Report the median too.
- SLAM = VO + map + loop closure (global consistency). MonoSLAM (EKF), KinectFusion (TSDF), SemanticFusion, SLAM++, QuadricSLAM.
- ORB-SLAM3: tracking, local mapping (local BA), loop & map merging (DBoW2, pose graph, full BA), Atlas. Relocalise = lost; loop = same map; merge = other map.

W8 Deep learning

- Task → output / supervision / loss / metric: cls (CE, acc); sem-seg (pixel CE, mIoU); det (cls + box, AP@IoU); inst-seg (mask AP); depth (L2, RMSE); retrieval (contrastive, R@K); obstacle (BCE, P/R at threshold).
- Backbone → embedding → head. Transfer ladder: feature reuse → linear probe → partial FT → full FT → scratch.
- Train / val (tune) / test (once). Loss ≠ metric.
- SSL: contrastive / triplet, MAE, DINO (student–teacher), CLIP.
- CNN: weight sharing, equivariant (not invariant). ViT: $N = HW/P^2$, [class] token, position embeddings, $\text{softmax}(QK^T/\sqrt{d})V$, residual. Less inductive bias → needs big data (ViT-H/14 88.55%).
- ↓threshold → ↑recall ↑FP; ↑threshold → ↑precision ↑FN. Softmax ≠ calibrated.

W9 Learned 3D

- Mono depth: ill-posed, global scale ambiguous. Supervision: direct GT / self-sup photometric (I_L(u,v) vs I_R(u−d,v)) / pseudo-labels (Depth Anything V2: synthetic teacher, 62M images).
- Depth map H×W×1 vs pointmap H×W×3. DUSt3R: pair → 2 pointmaps in cam-1 frame + confidence; up to scale; global alignment (σ_e, P_e, $\Pi\sigma = 1$); MASt3R adds descriptors and metric scale.
- Occ-Net f(x,h) → p_occ; DeepSDF → signed distance; NeRF f(x,y,z,θ,φ) → (σ, RGB), volume rendering, no GT geometry; 3DGS: mean, covariance, opacity, SH colour, splat. LERF/LangSplat + CLIP.
- Pretty renders ≠ validated occupancy.

W10 Detection

- Variable output count. Two-stage (anchors 3×3, RPN objectness, top-N, RoI pool, head); one-stage anchor (SSD, RetinaNet) / point (FCOS centerness, YOLO); DETR (queries, Hungarian, no NMS).
- Losses: CE/BCE/focal + L1/GIoU (+ objectness). Imbalance: focal, sampling, 1-to-1 matching.
- **IoU = I/(A + B − I)**. TP: IoU ≥ τ, right class, GT not already matched. P = TP/(TP+FP); R = TP/(TP+FN).
- AP = area under PR (monotone envelope, 101 pts); mAP = mean over classes (COCO also over IoU .5:.95). AP ≠ precision at the deployed threshold.
- Closed / open set / open vocabulary (CLIP, Grounding DINO); prompts matter; ≠ reasoning.
- DETR: 42.0 AP = FRCNN-FPN+; APL +7.8, APS −5.5; 100 queries; slow training.

W11 Seg & grasp

- Semantic (class per pixel) vs instance (mask per object). C logits/pixel; softmax (exclusive) vs sigmoid.
- Encoder–decoder; pooling / strided conv ↓, unpool / transposed conv ↑; U-Net skips keep boundaries. Mask IoU = $|P \cap G|/|P \cup G|$.
- Imbalance: all-background → high accuracy, obstacle IoU 0.
- Grasp planar (x,y,θ,w,q); 6-DoF (x,y,z,φ,θ,ψ,w,q). Generative (GG-CNN per pixel, detection-style) vs discriminative (Dex-Net, slow). Diffusion: grasp distribution. Task-aware: language (GraspMolmo, GraspVLA).
- Contact-GraspNet: contacts in scene cloud, 4-DoF (a, b, w); 17.7M sim grasps; 90.2% vs 80.4%.

W12 Place recognition

- VPR = image retrieval (not classification): loop closure, kidnapped robot, PnP candidates.
- Similarity matrix + threshold; single vs multi-session; challenges: viewpoint, illumination, season, dynamics, aliasing.
- P = TP/(TP+FP); R = TP/GTP; F1 = 2PR/(P+R); max recall @100% P; AUPRC; **R@K** = share of queries with a correct match in the top K.
- Global (compact, fast) vs local (robust, costly). BoW + TF-IDF t = (n_id/n_d)·log(N/n_i). NetVLAD; AnyLoc: DINOv2 + GeM/VLAD, domain vocabularies, up to 4× R@1.
- cos = $q^T a/(\|q\|\|a\|)$. Triplet / contrastive, hard negatives. Verify geometrically or with sequences (SeqSLAM).

W13 Failure & uncertainty

- OOD ($p_{train}(x) \approx 0$) vs open set (new class) vs open world (learn new). Covariate p(x); label p(y); concept p(y|x); novel class.
- Calibrated: n% confidence → n% correct. Bin: acc < conf → over-confident. ECE = $\Sigma(n_b/N)|acc - conf|$.
- Temperature: softmax(z/T), T > 1 softens; argmax unchanged; fit on validation.
- Epistemic (reducible with data; ensembles, MC dropout) vs aleatoric (irreducible; $L = \|y-\hat{y}\|^2/2\sigma^2 + \frac{1}{2}\log\sigma^2$). Entropy $H = -\Sigma p \log p$.
- OOD: max softmax, feature distance, energy E = −log Σ e^z. OOD ≠ failure.
- Selective: predict if r(x) < τ. Coverage = predicted/all; risk = wrong/predicted.
- Act on risk: abstain, slow down, re-observe, ask a human. No universal threshold.

Practice-exam answer key

1b 2a 3a 4c 5d 6d 7b 8c 9c 10a · 11b 12c 13b 14c 15d 16a 17a 18a 19d 20b · 21a 22d 23c 24c 25b 26d 27d 28d 29a 30d · 31b 32d 33a 34b 35c 36b 37c 38c 39a 40b

Practical results (optional, Phase 2)

The notebooks have not been run yet, so there are no results of yours to quote. Only the LightGlue demo, Week 5 and Week 6 notebooks contain instructor outputs. This section will be filled in Phase 2.